

E-Commerce Sales Insights Dashboard - Project Report

Executive Summary

This project analyzes e-commerce sales performance using Power BI. The dashboard helps management track revenue, profitability, customer behavior, product performance, and regional trends for better business decisions.

Project Objective

Design an interactive Power BI dashboard to monitor sales KPIs, identify growth opportunities, evaluate product performance, and improve operational efficiency.

Dataset Overview

The dataset contains order information, customer details, product categories, sales amounts, costs, profits, payment methods, shipping modes, delivery status, and customer segments.

Data Cleaning & Preparation

Activities include handling missing values, removing duplicates, standardizing categories, formatting dates, validating records, and creating calculated metrics such as Sales Amount, Profit Amount, and Profit Margin.

Key KPIs

Total Sales, Total Profit, Profit Margin %, Total Orders, Average Order Value (AOV), Revenue Growth %, Category Revenue, Region-wise Sales, Customer Contribution %, and Product Contribution %.

Dashboard Visualizations

KPI Cards, Sales Trend Line Chart, Category Performance Chart, Region-wise Sales Analysis, Top Products Chart, Profitability Analysis, Customer Segment Analysis, Payment Method Analysis, and Delivery Status Analysis.

Business Insights

Identify best-selling products, most profitable categories, top-performing regions, high-value customers, discount impact on profitability, seasonal sales trends, and delivery performance issues.

Business Recommendations

Increase focus on high-margin categories, optimize discounts, improve logistics in low-performing regions, strengthen customer retention strategies, and promote profitable products.

Tools & Technologies

Power BI, Power Query, DAX, Microsoft Excel, Data Analytics, Business Intelligence, Data Visualization, and GitHub.

Conclusion

The dashboard provides a centralized view of business performance and enables data-driven decision-making across sales, operations, marketing, and inventory management.

Project Deliverables

1. Power BI Dashboard (.pbix)
2. Sales Dataset (.xlsx)
3. Dashboard Screenshots
4. Project Report PDF
5. GitHub Repository
6. README Documentation